

- 23** The PDx1 protein is found in a wide range of animal species where it is regarded as essential for normal metabolism. In sand rats, the gene coding for the PDx1 protein has a much lower proportion of A and T nucleotides than *PDx1* genes from other animals.

Finding the complete DNA sequence of the *PDx1* gene in sand rats has been difficult. Sequencing involves splitting off nucleotides one at a time from single-stranded DNA.

What could account for the difficulty in finding the DNA sequence of the *PDx1* gene in sand rats?

- 1 The *PDx1* gene is present in only a small proportion of nuclei.
- 2 The *PDx1* gene is transcribed in only some cells.
- 3 The strength of the hydrogen bonding between the two strands of the *PDx1* gene is unusually high.

A 1, 2 and 3 **B** 1 only **C** 2 only **D** 3 only